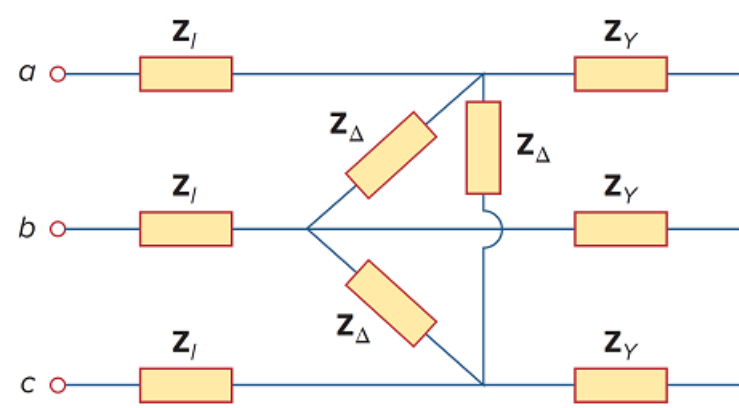


Problem

Refer to Fig. 12.48. Obtain the complex power absorbed by the combined loads.

Figure 12.48



Step-by-step solution

Step 1 of 1

$$S = 3 |I_p|^2 Z_p, \quad I_p = I_L \quad \text{For } Y\text{-Connected load}$$

$$S = (3) (13.66)^2 (7.812 - j2.047)$$

$$S = 4.373 - j1.145 \text{ KVA}.$$